ESE 2180 Project Two: Least-Squares Image Classification

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1 Project Information

This project will explore least squares methods for classification. The project is due on **November 3**. You may work in groups of up to **two**. You will turn in a lab report and source code.

2 Least Squares Classification

Classification is one of the fundamental problems in data science. The goal of classification is to correctly identify the "class" of a particular object. This could include identifying objects in an image or video frame, identifying the speaker of a particular audio file, or categorizing the type of music in an audio recording.

Mathematically, the classification problem can be stated as follows. Let $\mathcal{X}$ denote the set of possible inputs, and let $\mathcal{C}$ denote the set of classes. Let $f : \mathcal{X} \to \mathcal{C}$ be a function that maps each input to its class. The goal of classification is to construct a function $\tilde{f} : \mathcal{X} \to \mathcal{C}$ such that $\tilde{f}$ is "close" to f in some sense based on a set of training data $\{(x_1, y_1), \ldots, (x_N, y_N)\}$, where $x_i \in \mathcal{X}$ and $y_i = f(x_i)$.

In this project, we will explore a simple type of classifier, namely, the least squares classifier. Our discussion follows that in [1, Ch. 14]. For simplicity, we assume that $\mathcal{C} = \{-1, 1\}$, i.e., there are two possible labels. The idea of the least squares classifier is to choose

$$\tilde{f}(x) = \text{sign}(\theta_0 + \theta_1 f_1(x) + \cdots + \theta_M f_M(x)) \tag{1}$$

where $f_i : \mathcal{X} \to \mathbb{R}$ is a *feature* of the input x, θ_0 is an intercept (offset) term, $\theta_1, \ldots, \theta_M$ are coefficients, and the function sign is defined by

$$\text{sign}(z) = \begin{cases} 1, & z > 0 \\ 0, & z = 0 \\ -1, & z < 0 \end{cases}$$

The parameters θ are chosen by solving a least-squares problem

$$\text{minimize} \quad ||A\theta - y||_2 \tag{2}$$

where $\theta = (\theta_0, \theta_1, \ldots, \theta_M) \in \mathbb{R}^{M+1}$, $y = (y_1, \ldots, y_N) \in \mathbb{R}^N$, and the matrix $A \in \mathbb{R}^{N \times (M+1)}$ has a first column of all ones (corresponding to the intercept θ_0) and $A_{i,j+1} = f_j(x_i)$ for $j = 1, \ldots, M$. In all steps of this project, A has more rows than columns ($N > M + 1$) and the least-squares problem has a unique solution.

Clearly, the effectiveness of this approach will depend on the choice of features $f_1, \ldots, f_M$. As part of the project, you will explore how the choice of features impacts the classification accuracy.

The accuracy of the classifier is evaluated using test data. The test data consists of a collection of samples $\{(\hat{x}_1, \hat{y}_1), \ldots, (\hat{x}_R, \hat{y}_R)\}$ with $\hat{x}_i \in \mathcal{X}$, $\hat{y}_i \in \mathcal{C}$, and $\hat{y}_i = f(\hat{x}_i)$. We say that the learned classifier $\tilde{f}(x)$ has a classification error on sample $\hat{x}_i$ if $\tilde{f}(\hat{x}_i) \neq \hat{y}_i$. The *error rate* is the number of test samples with classification errors divided by the total number of test samples. The *false positive rate* is the number of test samples with $\tilde{f}(\hat{x}_i) = 1$ and $\hat{y}_i = -1$ divided by the number of test samples with $\hat{y}_i = -1$. The false negative rate is the number of samples with $\tilde{f}(\hat{x}_i) = -1$ and $\hat{y}_i = 1$ divided by the number of test samples with $\hat{y}_i = 1$.

3 Least Squares for Image Classification

We consider the problem of classifying images. An image can be represented as a vector $x \in \mathbb{R}^n$, where n is the number of pixels in the image and x_i represents the color intensity in the i-th pixel of the image. The class is a description of what is in the image.

In this project, you will work with the MNIST dataset [2]. MNIST is a collection of images, each of which contains a handwritten digit, i.e., '0', '1', ..., '9'. We will focus on determining whether each image is a 0 (corresponding to class 1) or one of the digits 1-9 (corresponding to class -1). The dataset contains four files:

- "train-images-idx3-ubyte": This file contains the images in the *training* dataset.
- "train-labels-idx1-ubyte": This file contains the labels in the *training* dataset.
- "t10k-images-idx3-ubyte": This file contains the images in the *test* dataset.
- "t10k-labels-idx1-ubyte": This file contains the labels in the *test* dataset.

You are provided a Matlab file "readMNIST" that can read the contents of these files to Matlab arrays. The command

```
[imgs, labels] = readMNIST('train-images-idx3-ubyte', 'train-labels-idx1-ubyte',...
 num_imgs,0);
```

reads the first `num_imgs` entries in the files 'train-images-idx3-ubyte" and "train-labels-idx1-ubyte" into the arrays `imgs` and `labels`, respectively. `images` is a three-dimensional array, with `images(i,j,k)` equal to the intensity of the pixel in the i-th row and j-th column of image k. `labels` is a one-dimensional array with `labels(k)` equal to the type of digit (from 0-9) in the k-th image. A similar command can be used to extract arrays of images and labels from the test dataset files.

The features in the image classification problem are equal to the pixel intensities that are nonzero in at least 600 images.

4 Project Assignment

Please complete the following steps:

1. Download the training and test datasets. Use the provided Matlab function to load the first 5000 images and labels of the training dataset into Matlab.

2. Identify the row and column indices of the pixels that have nonzero intensities in at least 600 of these 5000 training images. The intensities of these pixels will serve as the features $f_1, \ldots, f_M$ in all subsequent steps.

3. Construct the matrix A (including the column of ones for the intercept) and the label vector y as in Eq. (2), where $y_i = 1$ if the i-th image is a 0 and $y_i = -1$ otherwise. Solve the least-squares problem to obtain the parameter vector θ. You may use the Matlab backslash operator, `theta = A \ y`, to solve the least-squares problem. Create a plot showing the values of the entries $\theta_1, \ldots, \theta_M$ at the corresponding pixel locations, and report the value of the intercept θ_0.

4. Load the first 5000 images and labels of the test dataset, and extract the same features as in Step 2. Using these images, compute the error rate, false positive rate, and false negative rate of your classifier.

5. Repeat Steps 3–4, except use only the first 400 images and labels of the training dataset to train the classifier. Keep the same features that you identified in Step 2 and the same 5000 test images. How did the error rate and false positive/negative rates change?

6. In what follows, we will experiment with changing the feature set. From now on, use the first 5000 images and labels of the training dataset. Let M_0 denote the number of features that you identified in Step 2, and let M_1 denote the number of new features. Generate a matrix $R \in \mathbb{R}^{M_1 \times M_0}$ whose entries are 1 or -1 with equal probability, using the Matlab commands `rng(15); R = sign(randn(M1, M0));`. Call `rng(15)` every time before generating R, so that your results are reproducible. Letting $x_i \in \mathbb{R}^{M_0}$ denote the vector of features from Step 2 for the i-th image (without the

constant 1 corresponding to the intercept), the new feature vector is given by $\max\{Rx_i, 0\} \in \mathbb{R}^{M_1}$, where the maximum is taken entrywise. Apply the same matrix R to both the training and the test images.

7. Train a least-squares classifier (with an intercept θ_0) using the new features, i.e., with $M = M_1$ in Eq. (1). Here the column of ones for the intercept is appended *after* the transformation, so that $A \in \mathbb{R}^{N \times (M_1+1)}$. Compute its error rate, false positive rate, and false negative rate on the 5000 test images. Repeat this for $M_1 = 10, 20, 50, 100, 1000, 1500, 2000, 3000, 4000, 4500, 4600, 4700, 4800, 4900$, and plot the test error rate as a function of M_1. You may use the log scale for the x-axis via `semilogx`.

5 Deliverables

Please turn in your report and commented source code. The source code should be submitted as a .zip file. The report should address the following items:

1. **Problem setup.** Report the dimensions of the matrix A and the label vector y used to train the classifier in Step 3. Explain why appending a column of ones to A accounts for the intercept θ_0 in Eq. (1).

2. **Least-squares solution.** Write down the procedure for solving the least-squares problem. Under what condition on A is $A^\top A$ invertible, so that the least-squares solution is unique?

3. **Interpretation of θ.** Provide a plot of $\theta_1, \ldots, \theta_M$ at their corresponding pixel locations, and report the intercept θ_0. Explain how these parameters jointly determine the predicted class of an image. What does a positive or negative θ_j indicate about the corresponding pixel?

4. **Classifier performance.** Report the error rate, false positive rate, and false negative rate of the classifier from Step 4. Compare the false positive rate with the false negative rate, and explain the difference you observe.

5. **Effect of reduced training data.** Report the same performance metrics for the classifier trained on 400 images in Step 5, and compare them with those from Step 4. How and why does the performance change when less training data is used?

6. **Random features with thresholding.** Explain how the new feature vectors $\max\{Rx_i, 0\}$ are constructed. Is this transformation linear? Suppose instead that the new features were Rx_i, i.e., without the thresholding. How are these new features related to the original features x_i?

7. **Effect of the number of features.** Present your plot of the test error rate versus M_1 and describe the trend. Do more features always lead to better performance? Would increasing M_1 still help if the features were Rx_i instead of $\max\{Rx_i, 0\}$? Answer the question about using Rx_i by reasoning only; no additional code is needed.

References

[1] S. Boyd and L. Vandenberghe, *Introduction to applied linear algebra: vectors, matrices, and least squares*. Cambridge university press, 2018.

[2] Y. LeCun, "The MNIST dataset of handwritten digits," http://yann.lecun.com/exdb/mnist/.